

Matrix functions and stability

Multiplication overhead of cubic sign compositions

Difficulty: challenging **Importance:** interesting to the community**Status:** [Lean verified](#)

Rating rationale: Challenging because optimal cubic compositions must be compared with unrestricted evaluation programs; community impact comes from reusable matrix-sign iterations.

Lean verification — 22 September 2026

The full canonical uniform stage bound is formally proved for every multiplication budget and every real gap in $(0,1)$, including both small-budget endpoints. Seven exports establish actual register-program costs, genuine approximation errors and least feasible stage counts. They passed LeanCert kernel audits, two independent nonauthor final reviews, and [fresh isolated Linux Comparator/default-kernel verification](#) with rejection and sandbox controls.

[Proof and reviews](#) · [Immutable proof revision](#) · [Retained Linux evidence](#). Exposition: George Stepaniants; prior asymptotic order: Cheon–Kim–Kim; cubic construction: Chen–Chow. Formalization: Sidney Holden with OpenAI Codex assistance. This verifies the stated uniform asymptotic order; no novelty, exact optimum, same-budget optimality or external human peer review is claimed.

Resolution - uniform constant-factor asymptotic order

Resolution recorded 2026-09-12. The canonical asymptotic-order target is settled:

$$T_{\min}(m, \delta) = \Theta(m + 1) \quad (0 < \delta < 1),$$

with absolute constants uniform in the gap, including gaps depending on the multiplication budget. The [Theorem in Section 1](#) and [proof in Sections 2-5](#) give the explicit bounds

$$\left\lfloor \frac{m}{2} \right\rfloor \leq T_{\min}(m, \delta) \leq m \quad (m \geq 2), \quad T_{\min}(0, \delta) = T_{\min}(1, \delta) = 1.$$

Expository proof-note author: George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology, Pasadena, California, USA. [Proof PDF](#) · [Standalone TeX](#).

Prior work and exact scope. Cheon, Kim and Kim’s [ASIACRYPT 2020 work](#), especially Lemma 3 and its constant-factor complexity analysis, already yields uniform constant-factor order by combination with the classical degree bound. The optimized cubic is the earlier [Chen–Chow construction, equations \(3.3\)-\(3.6\)](#), also discussed in [Polar Express, Appendix F](#). The present self-contained note supplies the explicit comparison $T_{\min} \leq m$; it claims neither a new cubic iteration nor first discovery of constant-factor optimality. The exact smallest stage count, the optimal leading constant, and the stronger source question comparing the errors at the same multiplication budget remain unanswered.

Verification. A separate [independent Codex-agent mathematical and scope review](#) returned PASS for the theorem and the literal canonical asymptotic-order target. A second [independent source and attribution review](#) confirms the scope and prior-work distinctions. The [submission record](#) preserves both reviews, the original candidate and exact checks. Substantial AI assistance is disclosed. This is informal automated-agent review, not external human peer review or formal verification. No novelty or priority certification is asserted.

The original target, permanent ID, references and dated history remain below. The displayed ratings are retained as historical ratings of that target.

Context and notation

For $m \in \mathbb{N}_0$ and $0 < \delta < 1$, put

$$I_\delta = [-1, -\delta] \cup [\delta, 1].$$

Let $\mathcal{P}_m$ consist of real polynomials computed from $1, x$ by straight-line programs using at most m non-scalar multiplications; real linear combinations cost nothing. Define

$$E_m(\delta) = \inf_{p \in \mathcal{P}_m} \max_{x \in I_\delta} |p(x) - \text{sign}(x)|.$$

The arithmetic model concerns a single polynomial identity valid for matrices of every size.

Problem statement

Define

$$C_T(\delta) = \inf_{a_t, b_t \in \mathbb{R}} \max_{x \in I_\delta} |(q_T \circ \dots \circ q_1)(x) - \text{sign}(x)|, \quad q_t(x) = a_t x + b_t x^3.$$

Determine the asymptotic dependence on m, δ of

$$T_{\min}(m, \delta) = \inf\{T \in \mathbb{N}_0 : C_T(\delta) \leq E_m(\delta)\},$$

where the empty composition is x and $\inf \emptyset = +\infty$. Each stage costs at most two matrix products.

References and status evidence

Amsel et al., [Simons workshop report](#), §6.3, Problem 6.5. Rubensson, Jarlebring and Lorentzon, [degree-eight recursive expansion](#), §7, provides a June 2026 follow-up discussion; it does not settle this comparison.

Scope

MF-01 asks for an optimum value over general evaluation programs. This problem measures the price of a particular, reusable composition architecture; the two tasks are related but have different requested outputs.

Audit — 2026-09-10

Rechecked [Problem 6.5](#) and the [degree-eight paper's concluding discussion](#). The optimal composition comparison remains posed. Searches for cubic and recursive sign-expansion improvements found no resolution of this exact multiplication-overhead target.